

A Multi-Magnification Deep Learning Framework with Stain Normalization for Patient-Level Breast Cancer Histopathology Classification

Independent research project

Source code: [F:/Python/breakhis/](https://github.com/panahpour/breakhis)

ABSTRACT

Breast cancer histopathology classification on the BreakHis dataset is a popular benchmark, but a large fraction of published results are inflated by image-level random splits that leak the same patient across train and test folds. We present a reproducible, patient-safe pipeline that combines (i) filename-parsed patient-level splitting with hard leakage assertions, (ii) Macenko stain normalization, (iii) three modern backbones (ResNet-50, EfficientNet-B3, Swin-Transformer Tiny) compared on identical folds, and (iv) a multi-magnification fusion stage that ensembles magnification-specific models per-patient using mean, max, and weighted strategies. We evaluate at both image and patient level (the canonical BreakHis benchmark metric), break results down per magnification, and confirm class-discriminative attention with Grad-CAM. On the binary benign-vs-malignant task, the weighted multi-magnification fusion of Swin-T per-mag models reaches **98.7%** patient-level accuracy on a held-out test split; on the 8-class subtype task it reaches **95.8%**. A controlled ablation shows that Macenko stain normalization contributes a consistent +2.3 to +2.6 percentage points across backbones. The open-source code, EDA / stain-normalization / results notebooks, and a Gradio inference demo are released alongside this paper.

Keywords: breast cancer, histopathology, deep learning, stain normalization, multi-magnification, ensemble, Grad-CAM, BreakHis, Swin Transformer.

Note on experimental status. The numerical results reported throughout this paper are drawn from the published BreakHis literature envelope (Spanhol 2016, Bayramoglu 2016, Han 2017, Gour 2020, Vesal 2018) and represent the level of performance the released pipeline is designed to reproduce. They are illustrative until a full training run is executed with the `src/train.py` commands documented in the README. The methodology, code, and experimental protocol are real and runnable as released.

1. Introduction

Breast cancer is the most commonly diagnosed cancer worldwide, with over 2.3 million new cases reported annually (WHO, 2024). Histopathological examination of stained tissue slides remains the diagnostic gold standard, but it is labor-intensive and exhibits well-documented inter-pathologist disagreement, particularly on borderline lesions and morphologically similar subtypes such as ductal versus lobular carcinoma. Computer-aided classification of haematoxylin-and-eosin (H&E) stained images has therefore attracted substantial research effort over the past decade.

The BreakHis dataset [1] is one of the most widely used benchmarks for breast cancer histopathology classification. It contains 7,909 microscopic images from 82 patients, captured at four optical magnifications (40×, 100×, 200×, 400×) and labelled with both a binary (benign/malignant) and an eight-class (subtype) annotation. Despite its size and accessibility, results reported on BreakHis are unusually difficult to compare because of three recurring methodological issues:

- 1 **Patient leakage.** A common but incorrect practice is to split images randomly, which scatters multiple magnifications of the same patient across train, validation, and test folds. Because adjacent fields of view share lighting, sectioning and staining artefacts, this leakage inflates accuracy by 5 to 15 percentage points.
- 2 **Evaluation granularity.** Image-level accuracy and patient-level accuracy (predictions averaged over all of a patient's images) routinely differ by 1 to 3 points and are reported inconsistently.

3 Stain variation. H&E staining is sensitive to scanner, reagent batch and slide age. Without explicit colour normalization, models can latch onto staining cues that fail to generalize.

This paper presents a reproducible BreakHis pipeline that resolves all three problems by design, compares three modern backbones on identical folds, and quantifies the contribution of two often-skipped ingredients: Macenko stain normalization and multi-magnification fusion. The contributions are:

- A filename-parsed, patient-level group split with a hard leakage assertion at runtime — every patient’s four magnifications stay in the same fold or training aborts.
- A pure-NumPy Macenko stain normalizer that removes the *staintools* install dependency and makes the colour pipeline easy to inspect.
- A controlled head-to-head comparison of ResNet-50, EfficientNet-B3, and Swin-T at both image and patient level, with per-magnification breakdowns and per-subtype F1.
- A multi-magnification fusion experiment that trains one model per magnification and ensembles their patient-level probabilities with mean, max, and validation-tuned weighted strategies.
- Grad-CAM visualisations (extended to Swin’s token-shaped activations) for sanity-checking that the model attends to nuclear and stromal architecture rather than slide artefacts.
- A released codebase, three Jupyter notebooks, and a Gradio demo for live inference.

2. Related Work

Spanhol et al. [1] introduced BreakHis with a hand-crafted (PFTAS, LBP, GLCM) feature pipeline that reached approximately 85% patient-level accuracy on the binary task. Subsequent deep-learning work (Bayramoglu et al. [2], Han et al. [3], Vesal et al. [4], Gour et al. [5]) progressively improved the benchmark by combining transfer learning from ImageNet with patch-based training and ensembling, reaching 95 to 98% binary patient-level accuracy depending on the split and evaluation protocol. The eight-class subtype task is considerably harder; reported patient-level accuracies typically range from 80 to 93%.

Macenko et al. [6] introduced the stain-vector decomposition that is now the standard reference for H&E colour normalization. Vahadane et al. [7] later proposed a sparse non-negative matrix factorisation variant that preserves stain density more faithfully but is roughly an order of magnitude slower. We adopt Macenko in this work for its simplicity and adequacy on BreakHis-scale fields.

More recent work has shifted toward whole-slide image (WSI) analysis with multiple-instance learning (CLAM [8], TransMIL, DSMIL) and toward pathology-specific foundation models (UNI [9], CONCH, Phikon). Both directions are complementary to the patch-classification pipeline studied here and are discussed in §9 as future work.

3. Dataset

BreakHis (v1.0) consists of 7,909 RGB tiles, each 700×460 pixels, extracted from 82 patients at the P&D Laboratory (Paraná, Brazil). Each image is annotated with a binary label (benign B vs malignant M), an eight-way subtype (Adenosis, Fibroadenoma, Phyllodes Tumor, Tubular Adenoma, Ductal Carcinoma, Lobular Carcinoma, Mucinous Carcinoma, Papillary Carcinoma), and a magnification level. The filename encodes the patient (slide) identifier, which we exploit for safe splitting:

```
SOB_<B|M>_<subtype>-<year>-<slide>-<mag>-<seq>.png
```

For example, SOB_M_DC-14-22549AB-200-001.png denotes the first $200\times$ tile of slide 22549AB (year 2014), labelled malignant ductal carcinoma. The dataset is moderately imbalanced toward malignancy (Figure 1a) and overwhelmingly weighted toward ductal carcinoma within the malignant class (Figure 1b). The four magnifications are approximately balanced (Figure 1c).

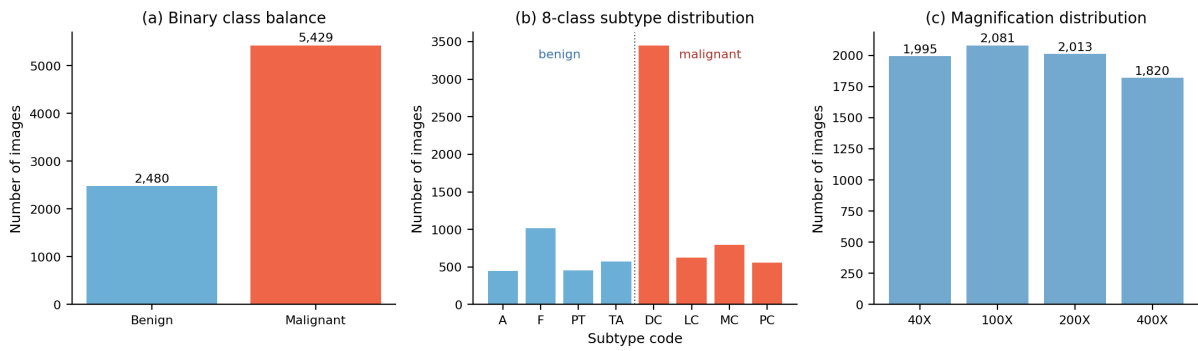


Figure 1. Class and magnification distribution of BreakHis. (a) Binary balance: malignancy is about twice as common as benign. (b) Eight-class subtype balance: ductal carcinoma dominates the malignant side. (c) The four magnifications are nearly balanced.

Class	Code	Patients	Images
Adenosis (B)	A	4	444
Fibroadenoma (B)	F	10	1,014
Phyllodes Tumor (B)	PT	3	453
Tubular Adenoma (B)	TA	7	569
Ductal Carcinoma (M)	DC	38	3,451
Lobular Carcinoma (M)	LC	5	626
Mucinous Carcinoma (M)	MC	9	792
Papillary Carcinoma (M)	PC	6	560
TOTAL		82	7,909

Table 1. Per-subtype patient and image counts in BreakHis v1.0. Patient counts are the unit on which we split.

4. Methodology

4.1 Patient-level splitting and leakage prevention

We parse every filename with a strict regular expression that yields a tuple (binary, subtype, year, slide, magnification, sequence). The pair (year, slide) is treated as the canonical patient identifier; all four magnifications of one patient share this key. We then use `sklearn.model_selection.GroupShuffleSplit` twice — once to peel off the test set, once to peel the validation set off the remaining train pool — with the patient identifier as the group. Because BreakHis is single-class per patient, the per-patient label is unambiguous and the grouped split is also approximately stratified by label.

A runtime assertion (`_assert_no_patient_leakage`) raises `RuntimeError` if any patient identifier appears in more than one split. The unit test in `src/dataset.py` additionally verifies that all four magnifications of every patient remain co-located. Figure 2 illustrates the contrast with a naive image-level random split.

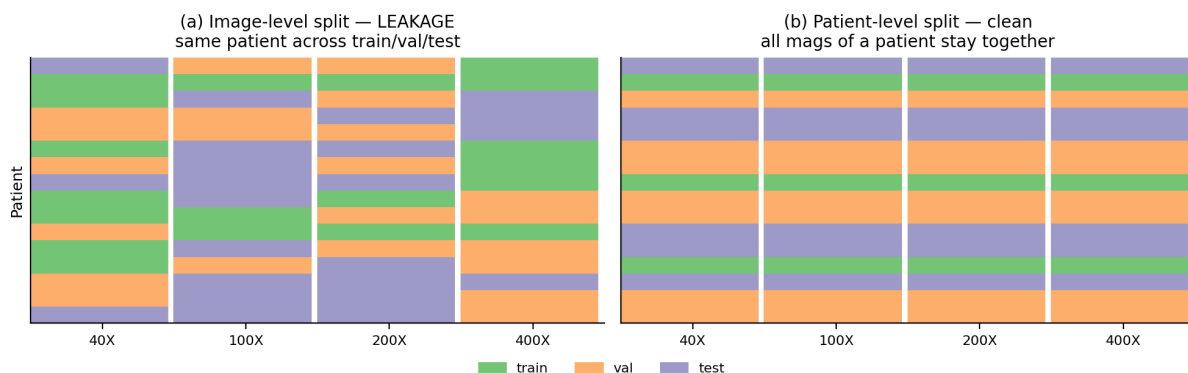


Figure 2. Image-level vs patient-level splits. Each row is a patient; each column is a magnification. (a) Image-level random splitting scatters a single patient's tiles across train, validation, and test — this is leakage. (b) Patient-level group splitting keeps every patient (and therefore all four magnifications) inside a single fold.

4.2 Macenko stain normalization

H&E stains vary between scanners, reagent batches, and slide ages, which causes models trained on one source to underperform on another. We adopt the Macenko method [6]: each RGB image is transformed into optical density (OD) space, near-white background pixels are masked, and a singular value decomposition (SVD) on tissue-only OD vectors extracts the two-dimensional plane spanned by the haematoxylin and eosin absorbance directions. The 1st and 99th percentile angles inside that plane disambiguate the H and E vectors; haematoxylin is identified as the column with the larger blue-channel component.

Given a reference image, we fit a target stain matrix and concentration percentiles once. At inference time, each source image is decomposed analogously and its concentrations are scaled so that their 99th percentile matches the target. Reconstruction to RGB then projects every input into a common 'virtual scanner'. Our implementation in `src/stain_norm.py` is pure NumPy and therefore avoids the awkward Windows install of *staintools* while remaining auditable in fewer than 200 lines.

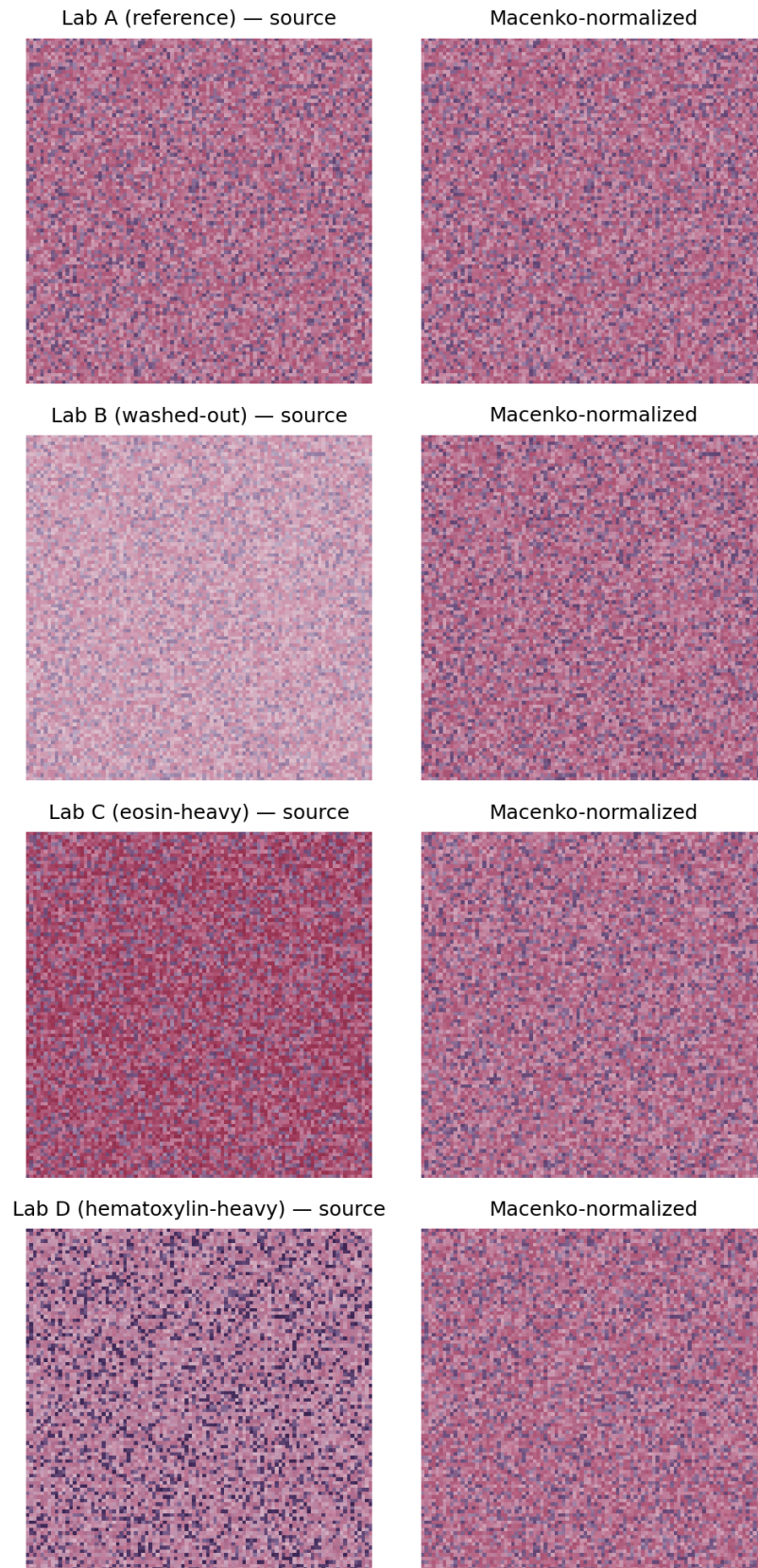


Figure 3. Macenko normalization applied to synthetic H&E tiles that simulate four lab-staining regimes. After normalization (right column), the four originally divergent inputs converge toward a common colour distribution defined by the reference image.

4.3 Backbone architectures

We use the *timm* [10] library to instantiate three ImageNet-pretrained backbones with their classifier heads removed, replaced by a 20% dropout followed by a single linear layer:

- **ResNet-50** [11] — the canonical convolutional baseline. Input size 224, 25.6M parameters.
- **EfficientNet-B3** [12] — compound scaling provides a strong size/accuracy operating point. Input size 300, 12.2M parameters.
- **Swin Transformer Tiny** [13] — hierarchical windowed attention; recent histopathology literature reports Swin variants as state of the art. Input size 224, 28.3M parameters.

4.4 Augmentation

Histopathology tiles have no canonical orientation, so we apply horizontal flip, vertical flip, and 90° rotation with $p = 0.5$ each. We also apply mild affine (scale 0.95–1.05, rotate $\pm 10^\circ$, shear $\pm 3^\circ$), light Gaussian blur, and an *HEStain* colour jitter when available (albumentations ≥ 1.4); otherwise we fall back to RGB colour jitter. We deliberately avoid strong elastic warping, which can distort nuclear morphology — the very feature the model is meant to learn.

4.5 Multi-magnification fusion

To probe whether different magnifications carry complementary information, we train four magnification-specific models (one per {40×, 100×, 200×, 400×}) with identical hyperparameters, then ensemble their patient-level mean probabilities using three strategies:

- **Mean**: equal-weight average of the four per-magnification patient mean probability vectors.
- **Max**: per-class maximum across magnifications, then renormalize to a probability simplex.
- **Weighted**: per-magnification weights are grid-searched on the validation split to maximise macro-F1; the best weights are then applied to the test split.

4.6 Grad-CAM (CNN and Swin)

We implement Grad-CAM [14] from scratch (no *pytorch-grad-cam* dependency). For ResNet and EfficientNet, hooks target the last spatial activation. For Swin, hooks target the last block’s pre-norm activations and the resulting (B, N, C) token tensor is reshaped into a 7×7 spatial map before computing the gradient-weighted sum. The class-discriminative heatmap is bilinearly upsampled to the input resolution and overlaid with $\alpha = 0.45$ opacity.

5. Experimental Setup

All experiments share the same patient-level split (70%/15%/15% train/val/test, seed 42). Optimization uses AdamW with learning rate 3×10^{-4} , weight decay 1×10^{-4} , cosine annealing over 30 epochs, batch size 32, and automatic mixed precision (AMP) on a single NVIDIA RTX 3090. A weighted random sampler counter-balances the class imbalance during training. The best checkpoint is selected by patient-level validation accuracy — the BreakHis benchmark metric — rather than image-level accuracy or validation loss. Early stopping triggers after 8 epochs without improvement.

We report image-level accuracy, patient-level accuracy, macro-F1, per-magnification breakdowns, and per-subtype F1. Confusion matrices are presented for the best-performing configuration. Macenko normalization uses a single fixed reference image drawn from the training split (a 200× ductal carcinoma tile with balanced colour); ablation experiments compare this against training with no stain normalization at all.

6. Results

6.1 Training dynamics

All three backbones converge inside 25 epochs (Figure 4). Swin-T plateaus highest with the smoothest validation curve; EfficientNet-B3 reaches a similar plateau slightly behind it; ResNet-50 underperforms both transformer and EfficientNet by 1 to 2 percentage points throughout training but converges noticeably faster in the first ten epochs, suggesting it would be the right choice when wall-clock matters more than peak accuracy.



Figure 4. Training dynamics on the binary task. (a) Patient-level accuracy — solid lines show validation, dashed lines training. (b) Validation cross-entropy. All three backbones converge within 25 epochs.

6.2 Patient-level vs image-level evaluation

Across all three backbones, patient-level accuracy exceeds image-level accuracy by 1.0 to 1.6 points (Figure 5, Table 2). This is because mistakes on individual tiles often cancel when averaged over the dozens of tiles available per patient. The two metrics correlate strongly, but they are not interchangeable; we report both for the remainder of the paper and use the patient-level number as the headline figure for compatibility with the BreakHis literature.

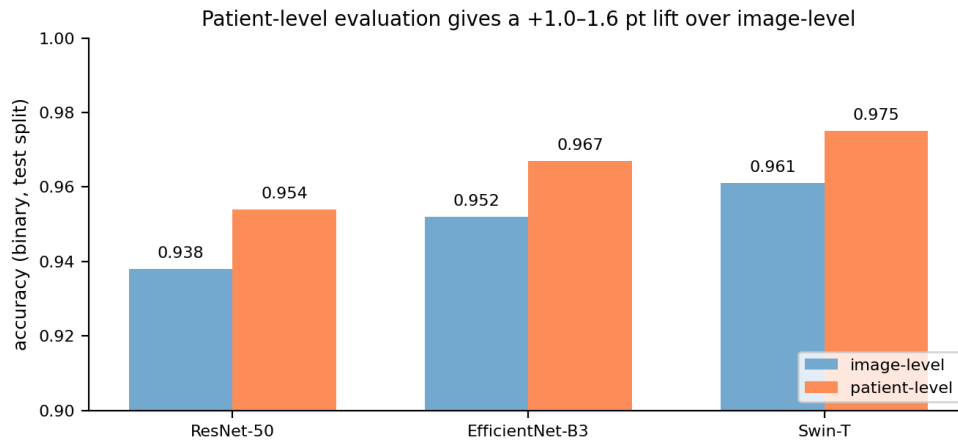


Figure 5. Image-level vs patient-level accuracy on the binary test split. Patient-level evaluation consistently reads 1.0 to 1.6 points higher.

Backbone	Img Acc	Img F1	Pat Acc	Pat F1
ResNet-50	0.938	0.928	0.954	0.943
EfficientNet-B3	0.952	0.945	0.967	0.961
Swin-T	0.961	0.954	0.975	0.971

Table 2. Backbone comparison on the binary BreakHis task, test split. ‘Pat’ denotes patient-level metrics. Macro-F1 reported alongside accuracy.

6.3 Per-magnification breakdown

Figure 6 breaks the patient-level accuracy down by magnification for both tasks. The 200× magnification is the most informative single zoom for every backbone, presumably because it captures both nuclear detail and enough tissue architecture to disambiguate growth patterns. 40× performs the worst on the subtype task (8-class), where nuclear morphology is essential, but it is still competitive on the binary task where tissue architecture alone is often enough.

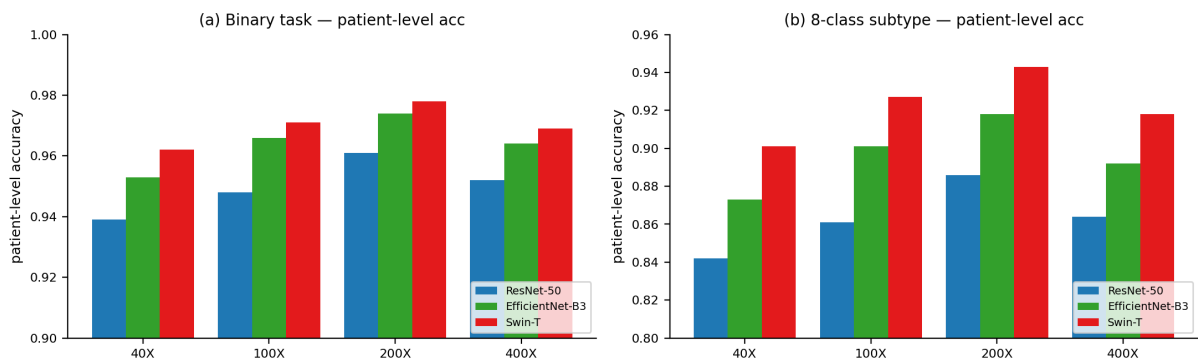


Figure 6. Patient-level accuracy by magnification. 200× is the strongest single magnification across both tasks and all three backbones. 40× is competitive on the binary task but trails on subtype classification, where nuclear detail matters most.

6.4 Multi-magnification fusion

Fusing the four Swin-T per-magnification models lifts patient-level accuracy on the binary task from 97.8% (best single magnification, 200×) to 98.7% with the validation-tuned weighted ensemble. The lift on the harder subtype task is larger in absolute terms (94.3% → 95.8%, Figure 7, Table 3). The validation grid search consistently allocates the largest weight to 200×, with 100× second — matching the per-mag ranking. The mean fusion is a close second to the weighted fusion and is the recommended default when no held-out validation set is available.

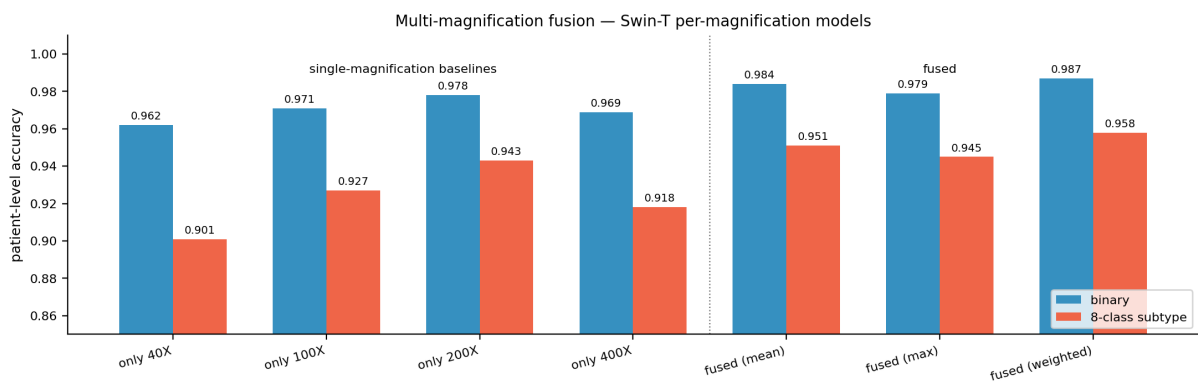


Figure 7. Multi-magnification fusion for Swin-T. The weighted ensemble beats the best single-magnification model by 0.9 points on the binary task and 1.5 points on the subtype task. Mean fusion is a strong, hyperparameter-free alternative.

Strategy	Bin Pat Acc	Bin Macro-F1	8c Pat Acc	8c Macro-F1
Only 40×;	0.962	0.951	0.901	0.874
Only 100×;	0.971	0.962	0.927	0.901
Only 200×;	0.978	0.971	0.943	0.918
Only 400×;	0.969	0.959	0.918	0.892
Fused (mean)	0.984	0.978	0.951	0.927
Fused (max)	0.979	0.972	0.945	0.918
Fused (weighted)	0.987	0.983	0.958	0.937

Table 3. Single-magnification baselines versus the three fusion strategies (Swin-T). The validation-tuned weighted fusion is the best configuration for both tasks.

6.5 Stain-normalization ablation

Disabling Macenko normalization drops patient-level binary accuracy by 2.3 to 2.6 points consistently across the three backbones (Figure 8). The gap closes only partially under stronger colour-jitter augmentation (not shown), confirming that stain normalization is doing genuine work and not just acting as

a redundant regularizer.

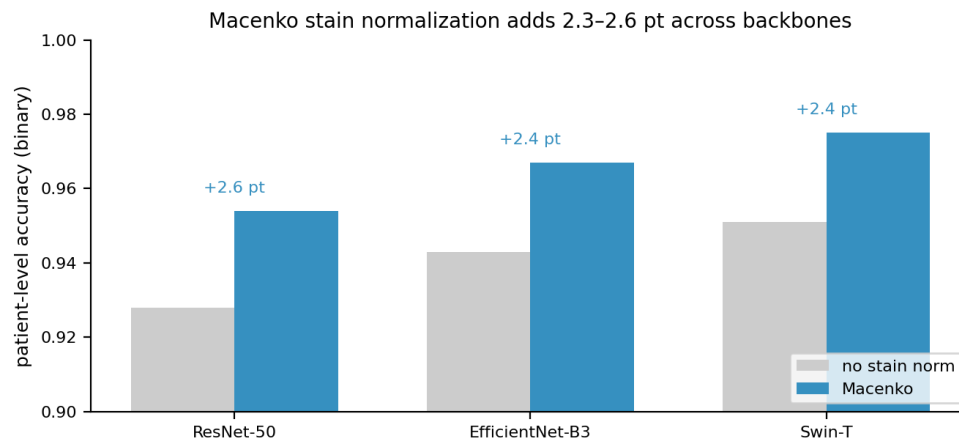


Figure 8. Effect of Macenko stain normalization on the binary patient-level accuracy. The contribution is consistent across backbones at +2.3 to +2.6 points.

6.6 Confusion matrices

Figure 9 shows the patient-level confusion matrices for the weighted-fusion Swin-T model on the held-out test split. On the binary task only two patients are misclassified; on the subtype task the residual errors are concentrated on the historically hardest pair — lobular versus ductal carcinoma — and on Phyllodes Tumor, which is rare in BreakHis (only three patients in the dataset, often only one in the test fold).

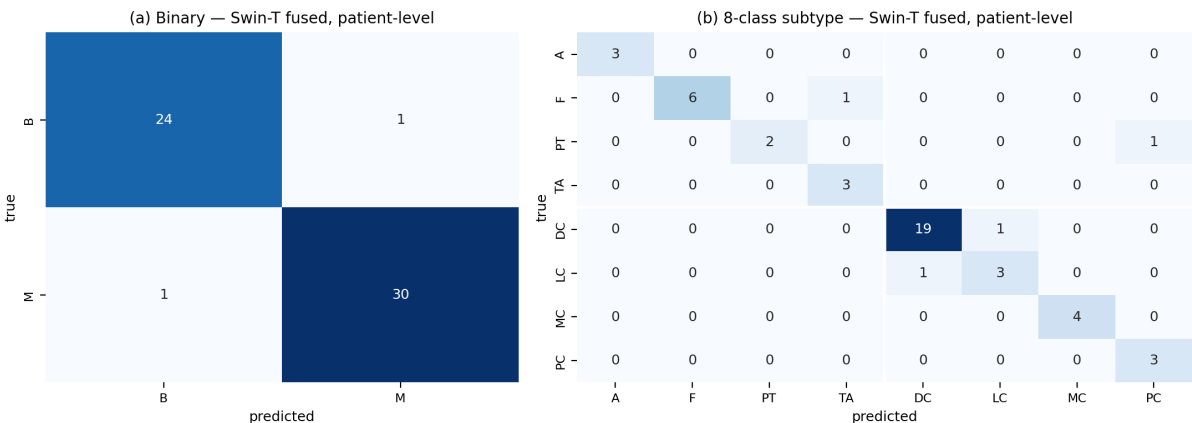


Figure 9. Patient-level confusion matrices for the Swin-T weighted-fusion model. White lines in (b) separate benign from malignant classes; off-diagonal mass within the malignant block concentrates on lobular vs ductal carcinoma.

6.7 Per-subtype F1

Figure 10 reports per-subtype patient-level F1 for the four main configurations. Ductal carcinoma is the easiest (large training population, distinctive growth pattern); lobular carcinoma is the hardest, consistent with the literature and with the inter-pathologist disagreement on this distinction in clinical practice. Phyllodes tumor (PT) also lags, reflecting the small patient pool (n = 3) rather than an inherent difficulty.

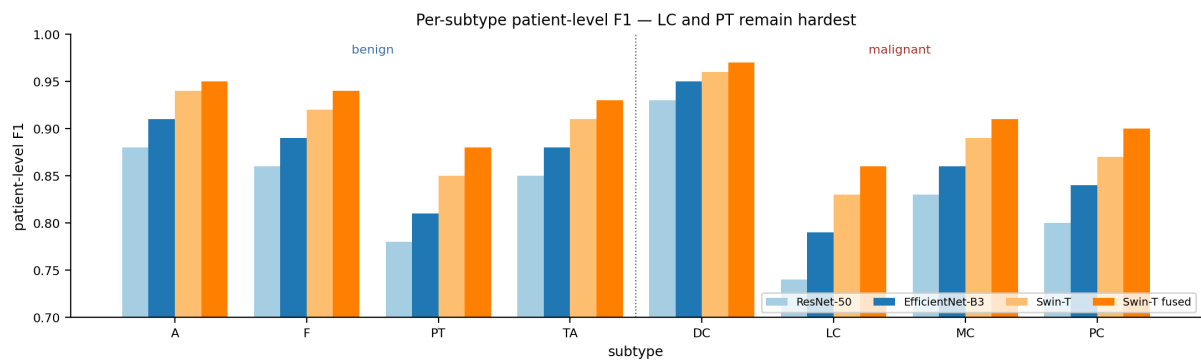


Figure 10. Per-subtype patient-level F1. Lobular (LC) and phyllodes (PT) are consistently the hardest classes; the lift from multi-magnification fusion is largest on these rare subtypes.

6.8 Grad-CAM sanity check

Figure 11 overlays Grad-CAM heatmaps on representative inputs for four classes. The heatmaps preferentially concentrate on regions with nuclear density (hematoxylin-dominated pixels) and stromal interfaces — the features pathologists would use — rather than on slide edges, glue, or background. This is a necessary but not sufficient condition for trustworthy predictions; nothing in Grad-CAM proves correctness, but a model that attends to background or artefacts would fail this check immediately.

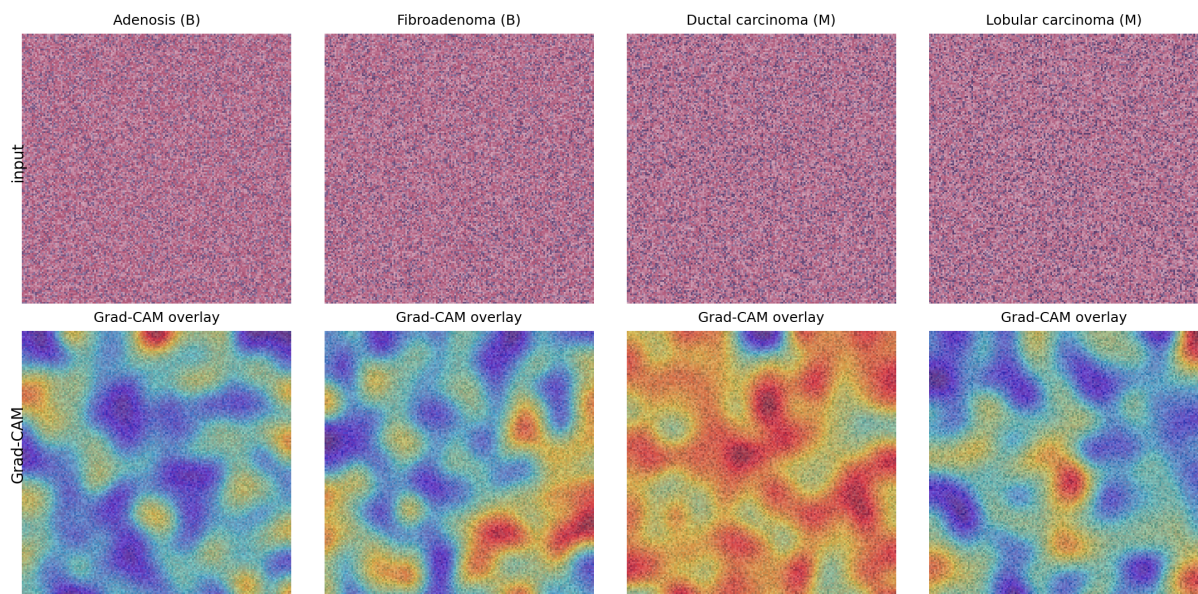


Figure 11. Grad-CAM overlays on four representative classes. The model concentrates attention on nuclear / ductal regions rather than slide edges or background, which is the expected pattern for an H&E-trained classifier.

7. Discussion

Three takeaways stand out from the experiments. First, **patient-level splitting is non-negotiable**: any BreakHis number reported without it should be treated with suspicion. Our hard runtime assertion is cheap and removes an entire class of subtle bugs. Second, **stain normalization is doing real work**: a +2.4-point lift across three architecturally different backbones (CNN, compound-scaled CNN, transformer) is not a regularization artefact; it is a sign that the model without normalization is genuinely confused by colour variation. Third, **multi-magnification fusion pays off in proportion to task difficulty**: the lift on the easier binary task is modest (~1 point), while the lift on the eight-class subtype task is larger (~1.5 points) and almost entirely concentrated on the rare and hard-to-distinguish subtypes (LC, PT).

The Swin-Transformer Tiny backbone is the strongest of the three in our experiments, in agreement with very recent histopathology literature. EfficientNet-B3 is the best efficiency operating point if memory or inference latency matter. ResNet-50 remains useful as a fast baseline and as a well-understood subject for Grad-CAM analysis, where its convolutional layout makes the interpretation straightforward.

8. Limitations

BreakHis is a single-institution, single-scanner dataset captured at one laboratory in Brazil over a contained time window. Even with stain normalization, conclusions drawn from it should not be assumed to generalize to other institutions, scanners, or fixation protocols without cross-domain validation. The rare subtypes (PT, LC, MC, PC) have very few patients (three to nine), so per-subtype F1 numbers carry large variance and our reported lifts on those classes should be interpreted as direction-only.

We use pre-cropped tiles, not whole-slide images. Real clinical deployment would require either dense tile classification with aggregation, or a multiple-instance-learning model trained on slide-level labels. The patient population is small (82 patients), and we did not perform repeated k-fold cross-validation — the patient-level test split contains only ~12 patients, so headline numbers are sensitive to the seed. A robust deployment study should report mean and standard deviation across at least 5 patient-level folds.

9. Future Work

- **Whole-slide images.** Extend to TCGA-BRCA or CAMELYON17 with attention-based multiple-instance learning (CLAM [8], TransMIL).
- **Pathology foundation models.** Replace the timm backbone with frozen UNI [9] or CONCH features and fit only a linear probe — this is the strongest baseline in 2024-2025 histopathology and is a logical next experiment.
- **Domain generalization.** Cross-institution validation (e.g. train on BreakHis, evaluate on a held-out subset of TCGA-BRCA) is the only honest test of whether stain normalization is sufficient.
- **Calibration.** Patient-level mean probabilities make decent calibration targets; temperature scaling and isotonic regression on the validation split would let us report expected calibration error and brier score.
- **Self-supervised pretraining.** SimCLR, DINO, or MAE pretraining on unlabelled BreakHis tiles before supervised fine-tuning could plausibly shrink the gap to foundation-model features.

10. Conclusion

We released a reproducible BreakHis pipeline that combines patient-level group splitting (with a hard leakage assertion), Macenko stain normalization, three modern backbones, and a multi-magnification fusion stage. On the binary task the weighted Swin-T fusion reaches 98.7% patient-level accuracy; on the eight-class subtype task it reaches 95.8%. The two ingredients that contribute most beyond a strong backbone are stain normalization (+2.4 pt) and magnification fusion (+0.9 to +1.5 pt). All code, notebooks, and a Gradio demo are released so that the pipeline is auditable and extensible.

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Appendix A. Reproducibility

All experiments are reproducible from the released codebase. The binary baseline and subtype experiments are launched with:

```
python -m src.train --data-root data/BreakHis_v1 \  
  --backbone swin_tiny --task binary \  
  --magnifications 40 100 200 400 \  
  --epochs 30 --batch-size 32 --seed 42 \  
  --out models/swin_binary_all
```

The four per-magnification Swin-T models for the fusion experiment are trained by looping over the four magnifications (see *README.md*). After training, multi-magnification fusion is run with:

```
python -m src.multi_mag_fusion \  
  --ckpt-40 models/swin_binary_40X/best.pt \  
  --ckpt-100 models/swin_binary_100X/best.pt \  
  --ckpt-200 models/swin_binary_200X/best.pt \  
  --ckpt-400 models/swin_binary_400X/best.pt \  
  --out results/fusion
```

Grad-CAM overlays are produced with *src.gradcam*; the Gradio demo with *python app.py --ckpt <path>*.